

Decrement gate

① Toffoli gate

$$\oplus = \boxed{\times} = \text{NOT Gate}$$

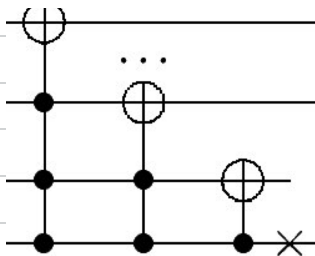
$\text{---}0\text{---} = \text{Turn on when } 0 = \text{Even switch}$

$\text{---}\bullet\text{---} = \text{Turn on when } 1 = \text{Odd switch}$



↑
when $|00\rangle \otimes |d\rangle \Rightarrow |d\rangle$ is flipped

② Decrement gate



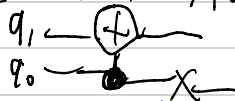
a) Increment gate

2bit Increment gate

$$\begin{cases} |00\rangle \rightarrow |01\rangle \\ |01\rangle \rightarrow |10\rangle \\ |10\rangle \rightarrow |11\rangle \\ |11\rangle \rightarrow |00\rangle \end{cases} \Rightarrow$$

$$|d\rangle \otimes |p\rangle$$

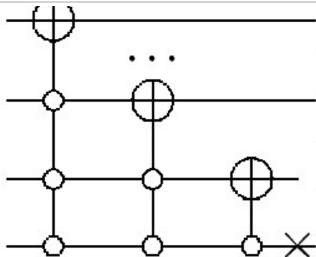
$|p\rangle$ always flipped
 $|d\rangle$ is flipped when $p \geq 1$



In general,

$$\Rightarrow |q_n q_{n-1} \dots q_i \dots q_0\rangle$$

q_0 always flip, $\uparrow$ Flips when $q_{i-1:0} = 1$



(b) Decrement gate

Decrement gate

$$\begin{cases} q_0 \text{ always flip} \\ q_i \text{ flips if } q_{i-1:0} = 1 \end{cases}$$